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Summary

Reference Number	34988
Lead Applicant	Mrs Lisa Edel
Research Title	Optimizing response to mechanical insufflation exsufflation to improve respiratory outcomes in children with Spinal Muscular Atrophy
Plain English Summary	**Background:** Spinal Muscular Atrophy (SMA) is a life-limiting condition, mostly affecting children, which weakens muscles, including those needed to move limbs, swallow, breathe and cough. As a result of weakened respiratory muscles, mucus builds up in the child's lungs, becoming thick and sticky, causing difficulty breathing. This leads to chest infections and hospital admissions affecting children's quality of life and life expectancy. Physiotherapists work with these children and their parents to prevent them from becoming unwell. One way these children can be helped is by daily use of a machine called a cough-assist, which clears mucus from the lungs. The cough-assist mimics a cough. Whilst the child temporarily wears a mask over their nose and mouth, air is blown into their lungs to help them take a big breath in. This is then followed by creating a vacuum to move mucus out. In SMA, we don't fully understand what happens to the child's airway whilst breathing, during coughing and when using cough-assist. This study will develop more understanding about optimisation of cough-assist settings, helping improve mucus clearance and will highlight benefits and issues relating to breathing affecting patient care. **Research aims and design:** There are four aims. 1. *Reviewing the current available literature to understand the current respiratory management for children with SMA and the future impact on their lives.* This will help ensure our study takes account of any new knowledge or developments. 2. *Understand patients', parents', and carers' views and experiences of day-to-day respiratory management.* Collaborating with SMA charity networks we will invite families to take part in online meetings, known as focus groups, to discuss key questions related to respiratory care. This will enable better understanding of issues which are important to children and families (These will be adapted for children so they can share their views). For the aims 3-4, we will invite children of all ages with a diagnosis of SMA under the care of Great Ormond Street Hospital to take part. 3. *Understand the practicality of using a small nasal camera to view what happens to airways of children with SMA when they breathe by themselves, use cough-assist and whether changing settings can improve coughing effectiveness.* Part a) Using a small (2mm) camera in a long thin tube, placed through the nose, we will look at the airway (this will not hurt). Two recordings will be made: one with and one without the cough-assist. Part b) We will then adjust the cough-assist settings (small changes to the size of the breath provided and to cough strength) and re-assess the airway and cough effectiveness. 4. *Part a) Assess breathing patterns of children when relaxed and when using cough-assists on initial and adjusted settings.* Patients will lie relaxed on a bed, breathing normally, with markers on their chest. Eight cameras pick up signals from the markers and convert the information into the child's breathing pattern. Using the original and adjusted cough-assist settings, differences in breathing patterns will be reviewed. We will ask children and families for their feedback about both new techniques (these will be changed for children so they can share their views). *Part b) Understand patients', parents' and carers' experiences of taking part in the study and its impact on respiratory care.* Discuss how useful the equipment and new tests are to patients, helping us understand strengths, challenges and areas for improvement.

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	Sharing results The study will be supported by a patient/parent steering group. The findings will be shared through parent support groups/charities and presentations at meetings and conferences (UK and internationally). Submissions will be made to peer-reviewed medical/physiotherapy journals.

Pre-submission

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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Pre-submission

1. Application Summary Information

Host organisation (which will administer any award)
Great Ormond Street Hospital For Children NHS Foundation Trust

Partner organisation	University College London

Research Title	Optimizing response to mechanical insufflation exsufflation to improve respiratory outcomes in children with Spinal Muscular Atrophy

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2025

Research Type	Other (please specify)

Please provide more details on your selection above.
An observational mixed methods research project

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2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2005	06/2008	BSc(Hons)	Physiotherapy	United Kingdom	University of East London	First Class
09/2002	06/2005	BSc(Hons)	Sport and Exercise Science	United Kingdom	University of Birmingham	2:2

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
02/2024	-	Band 7 Paediatric Respiratory Neuromuscular Clinical and Research Physiotherapist	Great Ormond Street Hospital NHS Foundation Trust	30
02/2016	02/2024	Band 7 Paediatric Respiratory Neuromuscular Physiotherapist	Great Ormond Street Hospital NHS Foundation Trust	10
07/2014	02/2016	Band 7 Paediatric Respiratory Physiotherapist	Royal London NHS Foundation Trust	0
07/2013	07/2014	Band 7 Paediatric Respiratory Physiotherapist	Evelina London	0
01/2010	07/2013	Band 6 Physiotherapist	Guys and St Thomas NHS Foundation Trust	0
01/2009	01/2010	Band 5 Physiotherapist	Guys and St Thomas NHS Foundation Trust	0
07/2008	01/2009	Band 5 Physiotherapist	Milton Keynes NHS Foundation Trust	0

Research Grants held

Publication Record
This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Childs, AM., Turner, C., Astin, R., Bianchi, S., Bourke, J., Cunningham, V., Edel, L., Edwards, C., Farrant, P., Heraghty, J., James, M., Massey, C., Messer, B., Michel Sodhi, J., Murphy, PB., Schiava, M., Thomas, A., Trucco, F. and Guglieri, M.. (2023) ‘Development of respiratory care guidelines for Duchenne muscular dystrophy in the UK: key recommendations for clinical practice.’, *Thorax*.
O'Connor, G., Edel, L., Raquq, S., Bowerman, M., Szmurlo, A., Simpson, Z., Hardy, I., Fewtrell, M. and Baranello, G.. (2022) ‘Open-labelled study to monitor the effect of an amino acid formula on symptom management in children with spinal muscular atrophy type I: The SMAAF pilot study.’, *Nutrition in clinical practice : official publication of the American Society for Parenteral and Enteral Nutrition*, 38(4).
LoMauro, A., Mastella, C., Scoto, M., Muntoni, F., Chan, E., Edel, L., Aliverti, A. and Baranello, G.. (2022) ‘Can Breathing Pattern Assessment Predict the Need of Ventilatory Support in Treated Infants with Spinal Muscular Atrophy Type 1?’, *American journal of respiratory and critical care medicine*, 206(1).

Pre-submission

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Edel, L., Grime, C., Robinson, V., Manzur, A., Abel, F., Munot, P., Ridout, D., Scoto, M., Muntoni, F. and Chan, E.. (2021) ‘A new respiratory scoring system for evaluation of respiratory outcomes in children with spinal muscular atrophy type1 (SMA1) on SMN enhancing drugs.’, *Neuromuscular disorders : NMD*, 31(4).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/11/2023	NIHR BRC	NIHR BRC GOSH Pre Doctoral Internship	Primary Investigator
01/11/2023	NIHR	PPIE funding award	Primary Investigator
01/10/2018	World Muscle Society	Lea Rose Spinal Muscular Atrophy Prize for the Best Presentation on SMA	Physiotherapist
01/10/2008	University of East London	EMS Prize for Best Performance in Clinical Placement	Physiotherapist

ORCID iD 0000-0002-9352-5388

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3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
PH88636 - HCPC

Please describe your research career to date
Following qualification as a physiotherapist in 2008 and gaining experience across a range of specialities, I found my passion for paediatric respiratory physiotherapy. I aim to optimise children's health and wellbeing, throughout their illness trajectory, building long-term clinical relationships with families, and educating on respiratory management. Working in specialist children's hospital services over the last 12 years, I expanded my experience, knowledge, and clinical expertise, developing a specific interest in respiratory physiotherapy in neuromuscular conditions. Respiratory compromise in this area is under-recognised and under researched, however it accounts for the majority of hospital admissions, and is the primary cause of mortality. In 2016, I was appointed to develop and lead the respiratory neuromuscular physiotherapy service. Care is provided in a variety of settings, including intensive care, inpatient wards and outpatients for children referred from across the United Kingdom. Through implementing effective interventions and bespoke care plans, this service enables longer healthier lives maximising quality of life. My clinical practice is focused on respiratory care from diagnosis through to transition to adult services, ensuring patients and families are supported throughout their journey. A key innovation in this role has been developing and leading the service for mechanical insufflation exsufflation (MIE). MIE is a device used by children with muscle weakness to help them produce an effective cough, clear mucus and prevent chest infections. Recognising the need to share expertise in this rapidly developing field, I have built and led a network of physiotherapists for respiratory care in neuromuscular conditions providing teaching and collaboration. I lecture at University College London MSc Physiotherapy on respiratory physiotherapy and airway clearance techniques. I work alongside Muscular Dystrophy UK and Spinal Muscular Atrophy UK charities to provide education, extending my reach to special interest groups, national and international meetings. I am invited to speak internationally on respiratory care for spinal muscular atrophy (SMA) to multi-disciplinary teams, further increasing opportunities for collaboration. My special interest in respiratory care for children with SMA developed in response to clinical need and recognition that main outcome assessments for drug trials focused on duration of ventilation for these children. This fails to capture critical aspects of respiratory management, including care needs at home. This fuels my determination to focus on all aspects of care optimising outcomes. My experience as a clinician and desire to improve care of children with neuromuscular disease, requires an integration of research and clinical knowledge. Critically for me is undertaking doctoral studies as the first stage of a clinical academic career. This foundation is key to my ambition to progress through clinical academia to develop robust evidence to continue to improve the lives of children with serious respiratory conditions.

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Research Experience

My interest in research developed into a passion, especially being part of a research active physiotherapy department and the neuromuscular service linked to the UCL Institute of Child Health, forming the Dubowitz Neuromuscular Centre of Excellence.

I am interested in respiratory outcome measures, ensuring meaningful patient findings as well as clinical relevance. In 2016/7, I developed the Great Ormond Street Respiratory Score for SMA1 outcome measure which monitors ventilation and dependence on respiratory physiotherapy. I presented this at the World Muscle Society Conference in 2018 and was awarded the Lea Rose Award for best SMA presentation. This work is published and since implementation has enabled better consistency of care with measurable respiratory function. I presented the longitudinal data at the World Muscle Society Conference in 2023.

I continue to develop the research aspect of respiratory management of neuromuscular patients, within a focused research position. This role aims to develop and improve national and international respiratory neuromuscular networks and form links with charity and advocacy groups. I continue to explore respiratory outcome measures to better monitor these patients. Alongside this work, I am the junior faculty representative for the NIHR-BRC Patient and Public Involvement and experience(PPIE) steering group.

I have developed international networks within the neuromuscular respiratory community, including North America, Europe, and the Middle East. I work on international standards of care projects and am involved in creating a global respiratory measure to gain consensus regarding best patient care.

My commitment to research is demonstrated through my successful application for a competitive NIHR-BRC funded internship at Great Ormond Street Hospital, which I am using to develop a robust research protocol and fellowship application to undertake doctoral study. The Doctoral Clinical Academic Fellowship(DCAF) is an essential step to consolidate, formalise and expand my research skills and knowledge, providing a foundation to embark on a clinical academic career. It will provide me with tools to develop my understanding of research design, methodology and enhance statistical analysis skills. It will facilitate focused time on PPIE ensuring patient focused outcomes, improving physiotherapy care and treatment benefiting children with neuromuscular conditions.

Teaching and training

Education is a key component of my role and as I develop to become a leader in my field. I provide training to other professionals and students. I am committed to developing the next generation of physiotherapists and want to ensure high quality knowledge translation and supervision. They should be able to question practice and progress as a direct result. It is vital to remain up to date with current research and teach the importance of evidence-informed practice. I educate peers on the importance of data collection, conducting audits, service evaluations and quality improvement projects. My role and skill set have enabled me to teach at a senior level on national and international stages and I hope to continue to educate and develop clinicians as I further my own clinical academic career.

I am a highly motivated clinical specialist respiratory neuromuscular physiotherapist and my career has enabled me to become a recognised expert in my field, develop highly specialised clinical skills, lead on service developments, teach internationally, participate in and create my own research and remain driven to achieve in a clinical academic career. With a demonstrated commitment to research activity and a drive to improve, I am well placed to undertake an NIHR funded doctoral fellowship.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the

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assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

Not applicable

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4. Plain English Summary of Research

Background:

Spinal Muscular Atrophy (SMA) is a life-limiting condition, mostly affecting children, which weakens muscles, including those needed to move limbs, swallow, breathe and cough.

As a result of weakened respiratory muscles, mucus builds up in the child's lungs, becoming thick and sticky, causing difficulty breathing. This leads to chest infections and hospital admissions affecting children's quality of life and life expectancy. Physiotherapists work with these children and their parents to prevent them from becoming unwell. One way these children can be helped is by daily use of a machine called a cough-assist, which clears mucus from the lungs.

The cough-assist mimics a cough. Whilst the child temporarily wears a mask over their nose and mouth, air is blown into their lungs to help them take a big breath in. This is then followed by creating a vacuum to move mucus out.

In SMA, we don't fully understand what happens to the child's airway whilst breathing, during coughing and when using cough-assist.

This study will develop more understanding about optimisation of cough-assist settings, helping improve mucus clearance and will highlight benefits and issues relating to breathing affecting patient care.

Research aims and design:

There are four aims.

1. *Reviewing the current available literature to understand the current respiratory management for children with SMA and the future impact on their lives.*
   This will help ensure our study takes account of any new knowledge or developments.
2. *Understand patients', parents', and carers' views and experiences of day-to-day respiratory management.*
   Collaborating with SMA charity networks we will invite families to take part in online meetings, known as focus groups, to discuss key questions related to respiratory care. This will enable better understanding of issues which are important to children and families (These will be adapted for children so they can share their views).

For the aims 3-4, we will invite children of all ages with a diagnosis of SMA under the care of Great Ormond Street Hospital to take part.

3. *Understand the practicality of using a small nasal camera to view what happens to airways of children with SMA when they breathe by themselves, use cough-assist and whether changing settings can improve coughing effectiveness.*
   Part a) Using a small (2mm) camera in a long thin tube, placed through the nose, we will look at the airway (this will not hurt). Two recordings will be made: one with and one without the cough-assist.

   Part b) We will then adjust the cough-assist settings (small changes to the size of the breath provided and to cough strength) and re-assess the airway and cough effectiveness.
4. *Part a) Assess breathing patterns of children when relaxed and when using cough-assists on initial and adjusted settings.*
   Patients will lie relaxed on a bed, breathing normally, with markers on their chest. Eight cameras pick up signals from the markers and convert the information into the child's breathing pattern. Using the original and adjusted cough-assist settings, differences in breathing patterns will be reviewed. We will ask children and families for their feedback about both new techniques (these will be changed for children so they can share their views).

   Part b) Understand patients', parents' and carers' experiences of taking part in the study and its impact on respiratory care.

   Discuss how useful the equipment and new tests are to patients, helping us understand strengths, challenges and areas for improvement.

Sharing results

The study will be supported by a patient/parent steering group. The findings will be shared through parent support

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groups/charities and presentations at meetings and conferences (UK and internationally). Submissions will be made to peer-reviewed medical/physiotherapy journals.

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5. Scientific Abstract

Background

Spinal Muscular Atrophy(SMA) is a life-limiting condition affecting children's ability to move their limbs, swallow, breathe and cough effectively, which impacts on quality of life and life expectancy. This leads to increased thick and sticky secretions, causing chest infections and increased hospital admissions. Children with SMA spend approximately 20% of their lives in hospital, at an average cost of £2.2M(20). Chest Physiotherapists utilise devices called cough-assists(MIE) to prevent infections. These devices mimic a cough by delivering positive pressure to encourage deep inhalation followed by a rapid switch to negative pressure to stimulate a cough. Chest Physiotherapists set MIE on appropriate settings for the child to use in the community.

SMA is linked to a secretion build up, however, there is limited understanding of what happens to the child's airway and breathing during coughing and using the MIE. The main concern is potential airway collapse and MIE being ineffective. It is vital to have a better understanding of the airway response and how families and children feel about the respiratory aspect of their care.

Research Question

Optimizing response to mechanical insufflation exsufflation to improve respiratory outcomes in children with Spinal Muscular Atrophy

Aims

1: Investigate reasons for variability in effectiveness and tolerance of children to MIE.

2: Gain an understanding of day-to-day respiratory management for children with SMA, identifying key challenges, strengths and areas for improvement.

Objectives

1) Complete a comprehensive systematic review of the literature using PRISMA-P protocol.

2) Explore patients and parents experiences about their respiratory management and the impact on children's daily lives.

3a) Assess feasibility of Transnasal Fibreoptic Laryngoscope for airway assessment and during cough assist with children with SMA and describe and evaluate the upper airway response patterns with MIE.

3b) Evaluate whether changes in MIE settings can alter the response patterns seen in the upper airway.

4a) Assess changes in spontaneous respiratory patterns utilising opto-electronic plethysmography with optimised MIE settings.

4b) Understand patients, parents and carers acceptability of new assessment techniques and their views on the contribution to respiratory care outcomes.

Method

An observational cohort mixed methods study consisting of 4 work packages(WP).The study will be supported by a parent steering group.

Following the systematic review(WP1), focus meetings with SMA families will be undertaken to gain views on current respiratory management from across the United Kingdom(WP2). Thereafter, 50 patients will participate in two innovative respiratory assessment focused on response to MIE and respiratory function(WP3a/b/WP4a). Statistical analysis will be conducted. Finally, semi-structured interviews will take place to gain information on the new assessments(WP4b). Data synthesis will confirm challenges, strengths and areas for improvement.

Timelines for delivery

WP1:Months 1-6

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WP2:Months 5-14

WP3 and 4:Months 14-21

Anticipated impact and dissemination

The impact focuses on enabling improved secretion management for patients, specifically improving health and wellbeing of patients and families and optimising NHS resources.

Results will be shared with parent support groups/charities as well as through presentations at meetings and conferences, across the UK and internationally. It will be shared through submission to appropriate peer-reviewed medical/physiotherapy journals.

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6. Detailed Research Plan

What is the problem being addressed?

Spinal Muscular Atrophy(SMA) is an autosomal recessive neuromuscular disorder caused by the deletion of survival of motor neuron protein 1(*SMN1*), responsible for 90% of SMN production in healthy individuals(1). Another gene, survival motor neuron 2(*SMN2*) produces insufficient protein to maintain muscle function. Due to lack of SMN protein, SMA is characterised by degeneration of spinal cord anterior horn cells leading to muscle atrophy and proximal muscle weakness(2).

SMA affects all ethnicities, with higher mutation rates in Caucasian and Asian populations (excluding Asian Indian), and an incidence rate of 1 in 10,000 live births(3). SMA is divided into subtypes based on severity and age of onset, with number of SMN2 copies correlating with severity(4-6).

There are 5 different types of SMA; the most severe SMA0 to the adult form SMA4. Infants with SMA0 are born with cardiac and respiratory complications(7), rarely surviving the first few months of life.

SMA1 is the most common form, presenting in the first six months of life with progressive muscle weakness and bulbar dysfunction with resultant chest infections and respiratory failure. Respiratory failure is the major cause of morbidity and mortality. SMA1 is sub-grouped into types A, B, or C, with type 1A being the most severe(8). Historically, children with SMA1 died before 2 years old. Over the past 15–20 years, standards of care, supportive therapies and disease modifying treatments(DMT) have driven an interventional approach and have improved life expectancy(9,10). SMA1 has shifted to a chronic life-limiting condition(11) with children living with the disease, attending school and participating in the community. SMA2 and SMA3 have fewer respiratory complications, but respiratory compromise may occur.

Children with SMA1, 2 or 3 struggle with secretion clearance due to an ineffective cough, resulting in mucus plugging, impaired mucociliary clearance and lung collapse(12). Effective secretion management and respiratory physiotherapy play a crucial role in preventing respiratory infections, reducing hospitalisations, and improving life expectancy(13). These patients require regular reviews with a specialist physiotherapist and individualised treatment programmes, including one specific treatment; mechanical insufflation-exsufflation(MIE), commonly called a cough-assist machine(14). This device mimics a cough by utilising positive pressure to encourage deep inhalation (insufflation), followed by a rapid switch to negative pressure (exsufflation) creating a vacuum and subsequently increasing expiratory flow rate. This moves secretions higher in the oropharynx allowing effective clearance. MIE is recommended by international standards of care as a vital treatment for patients with SMA(15).

Despite DMT targeting motor and respiratory function for SMA, optimizing respiratory management remains essential to patient care. MIE is used worldwide for SMA patients(12). Determining optimal settings for MIE devices remains a discussion point. Currently, clinicians rely on observation and patient feedback to establish appropriate pressures, lacking clear evidence based guidance(16).

SMA patients across all subtypes can have elements of bulbar dysfunction, with the highest percentage being in the SMA1 category(11). SMA patients with bulbar dysfunction face additional challenges in utilizing MIE due to degeneration in corticobulbar and brainstem pathways(17). Research on upper airway management and response to MIE in amyotrophic lateral sclerosis(ALS), shows therapeutic pressures with MIE may cause a disadvantageous laryngeal response, preventing the lungs from filling with air and consequently reducing cough efficacy. Trans-nasal fibreoptic laryngoscopy(TFL) has been used in this patient population to view the upper airway and assess the effect of MIE. It monitored responses throughout all phases of coughing(18). However, upper airway response and resulting lung function with MIE in children with SMA remains unknown.

Further research is necessary to determine the upper airway response to MIE in SMA and develop evidence-based pressure settings for MIE.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or patients and health and care services?

This has been identified by the James Lind Alliance, under the SMA Top 10 priorities, number 5 (What is the best physiotherapy approach for people with SMA? Does it need to be tailored to the individual (e.g. according to age and type of SMA)? Does it need to change during treatment with one of the new drugs?).

- *Improving respiratory outcomes:* By studying the upper airway during MIE using TFL, the research aims to determine the most effective delivery of pressures, optimising lung expansion utilising opto-electronic plethysmography(OEP) and reducing risk of lung collapse. This leads to improved respiratory function,

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decreased chest infections, fewer hospital admissions, directly impacting morbidity and mortality in SMA patients.

- *Reducing hospital admissions:* Despite DMT, hospitalisations remain high due to respiratory complications in SMA patients. Approximately 20% of a SMA1 child's life is spent in hospital, with an estimated lifetime cost of £2.2M(20). Similarly, in the United States, each admission for a child with SMA1 costs appropriately $50,190 and they require an average of 4.2 hospitalisations annually(21). There is a large physiological, psychological, educational and economic burden to the patient and family(22,23).
- *Improving patient/carer wellbeing:* Limited information exists regarding respiratory burden of care for patients and families. Recent studies investigated user perspective for MIE in relation to effectiveness(16). However, there is no validated measure for respiratory burden in neuromuscular conditions. This research aims to identify key components affecting experiences of patients and families, aiming to improve management and overall wellbeing in terms of respiratory care.
- *Supporting development of care pathways.* Current standards of care(15) highlight importance of MIE for SMA patients(2,12,13). Limited data is available to guide optimal settings and frequency of use. Additionally, access to funding for these devices is variable across the United Kingdom. This research will guide device set-up and create clinical guidelines, which would be of benefit to clinicians and patients.

Review of existing evidence

Optimal respiratory management is critical as respiratory complications remain the primary cause of mortality and morbidity, despite disease modifying treatments(8).

Use of TFL

Airway clearance for patients with SMA is highly complex and has a high impact. The existing literature reviewing the upper airway in adult ALS(17) and a single case review in a child with SMA1(24), do not enable conclusions to be made regarding a larger SMA cohort.

MIE is not effective for all patients(17). Poor bulbar control may cause obstruction to airflow through the larynx. The larynx consists of 3 main layers along with mucosa, ligaments and muscles that are responsible for abduction (opening) and adduction (closing) of the airway and this co-ordination is vital for effective use of MIE. Literature in adults with ALS suggests individualised MIE settings are important, highlighting treatment failure is likely due to laryngeal adduction during insufflation(17,18). This response precludes air filling of the lungs during insufflation, causing discomfort and subsequent inefficient exsufflation. The clinical translation to SMA remains unclear. Utilising TFL is the gold standard test for visualising the upper airway(25), enabling separating out of the larynx into supraglottic, glottic and hypopharyngeal, and documentation of sequencing of airway abduction and adduction. It is well tolerated and used successfully(26).

Discussion on TFL during patient and public involvement(PPIE) activities, highlighted concern regarding its use. However, when the size was highlighted as being smaller than suction catheters used day-to-day, it was then acceptable. The patients and parents suggested including patient videos and step-by-step guides to help children understand the test. This will form part of the proposal.

Use of OEP

Lung function assessments in SMA1 children are challenging as it involves invasive or volitional tests. SMA1 patients often have more rapid and shallow breathing with paradoxical thoracoabdominal movements and bell shaped chests(27). OEP is utilised by researchers to investigate tidal and compartmental chest wall volumes. It measures quiet breathing, inspiratory capacity and voluntary cough. Advantages of utilising OEP are it can be performed in young infants as it does not require cooperation. Use in this population is particularly suitable due to a lack of reliable outcome measures. SMA children have breathing asynchrony and more restrictive chest walls(28). OEP is safe and well tolerated by patients of all ages (28). Research with other neuromuscular diseases show a single treatment with MIE can provide beneficial changes to the breathing pattern, although there were no changes in lung volume recruitment or cough peak flow(29).

Patient experience

Many questionnaires focus on quality of life, however, these are specific to other conditions, and are limited in the respiratory neuromuscular paediatric population. Alderhey Childrens hospital have validated a respiratory symptom questionnaire for children with neuro-disability(30), which contains elements relevant to children with SMA.

There is a need to improve the respiratory management of children with SMA and explore applications and

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mechanisms of airway clearance with MIE and its impact on lung function.

Research Question

Optimizing response to mechanical insufflation exsufflation to improve respiratory outcomes in children with Spinal Muscular Atrophy.

Aims:

1: Understand reasons for variability in effectiveness and tolerance of children to MIE.

2: Gain an understanding of day-to-day respiratory management for children with SMA, identifying key challenges, strengths and areas for improvement.

Objectives:

- Complete a comprehensive systematic review of literature aiming to answer 'what happens to upper airways and lung volumes in children with SMA (Work Package 1(WP1))
- To explore experiences of respiratory management as perceived by patients' and parents' in order to inform assessments impacting children's daily lives (Work Package 2(WP2))
- To determine the feasibility of TFL for airway assessment with children with SMA, describing and evaluating upper airway response patterns with MIE (Work Package 3a(WP3a))
- Evaluate whether changes in MIE settings alter response patterns seen in the upper airway (Work Package 3b(WP3b))
- Utilising OEP, assess changes in spontaneous respiratory patterns whilst using optimised MIE settings (Work Package 4a(WP4a))
- Understand patients', parents' and carers' views, tolerance and acceptance of new assessment techniques (Work Package 4b(WP4b))

Project Plan

An observational cohort mixed methods study has been designed to investigate upper airway response to MIE in children with SMA.

This study includes 4 work packages(WP). Time frames are included in the GANTT chart (see supporting documents). PPIE will occur throughout all work packages.

The development of this project was supported by the NIHR Research Support Service.

See attached for a schematic of the study design (appendix 2).

WP1: Systematic literature review

Using PRISMA-P protocol, literature related to SMA respiratory management, trans-nasal flexible laryngoscopy and opto-electronic plethysmography will be systematically reviewed. The question aiming to be answered is:

What happens to upper airways and lung volumes in children with SMA during MIE?

Study types will include but not be restricted to, randomised control trials, cohort studies and case studies. A range of databases including MEDLINE, CINAHL, EMBASE, EMCARE, SCOPUS, COCHRANE and web of science will be searched with full text articles obtained. Reference lists will also be searched.

Search strategies will be developed with an information specialist. Terms would include;

((“Spinal muscular atrophy”OR“neuromuscular disease*”) AND ("respiratory management"OR"respiratory physiotherap*"OR“airway clearance technique*”) OR (“mechanical insufflation exsufflation”) OR (“trans-nasal fibreoptic laryngoscopy”) OR (“opto-electronic plethysmography”) OR airway OR lung volume)

Search strategies will be limited by age(0-18years), written or translated to English and date of publication (2010 onwards).

Timeline-Months 1-6

WP2: Patient and Parent perspective of respiratory management

Pre-submission

Qualitative research, to better understand patient and parent experience related to respiratory management, will be completed for children and families with SMA.

- Patients and parents from national SMA networks and those captured via advertising through charity networks, will be asked to take part in online focus groups regarding respiratory management. This enables data collection and feedback from patients from across the UK, ensuring a diverse and representative sample group. Variation in standards of care exist throughout the United Kingdom and it is important to capture that impact on respiratory health.
- Utilising the Liverpool Respiratory Symptom Questionnaire and questions linked to clinical respiratory management gained from collaborative work with the PPIE group and clinicians involved in SMA care, a topic guide will be developed.
- Informed consent will be sought from parents and assent from children under 16 years of age.
- Focus groups will be facilitated to enable discussion with families, to share and compare experiences of respiratory management, collecting data to achieve a comprehensive understanding of respiratory management from the patient's perspective. We will offer children the opportunity to take part in child-focused group (with or without parental presence) where data collection approaches will reflect age and developmental stage.
- Facilitators will introduce the topic. Participants will discuss issues important to their experiences. Facilitators will encourage discussion allowing different and similar experiences to be voiced, giving prompts and clarification when required.
- Utilising a range of approaches, views of the family and child will be captured. For children, this includes picture cards/talking mats and emoji scales, recognising younger children may not feel confident in joining open ended question discussions.

Inclusion Criteria: Patients and families within the UK SMA reach network are eligible for completing the focus groups regarding their experience and understanding of respiratory management.

Every effort will be made to include all patient's and families from a diverse population, including offering interpreters.

Focus groups will last approximately 1 hour and will be recorded via MS Teams, with participants permission. It will be limited to 6 participants per group. This may include patients with a family member, which would account for 1 participant in this scenario. We anticipate information power being achieved by recruiting 24-30 participants, for this part of the study. We will pursue snowball sampling as data collection progresses to ensure a diverse group representative of the population of children with SMA in the UK. Data will be transcribed verbatim for analysis.

Data Analysis

Using NVIVO software to store and organise data, inductive thematic analysis will be used to analyse data collected from focus groups. Data collection and analysis will occur concurrently through an iterative process enabling data already collected to inform subsequent data collection. This constant comparative methodology will extend to thematic analysis of each data set until information power is achieved(31).

Once themes are highlighted, some will help guide prompts for WP4b.

Timeline-Months 5-14

Power Calculation for WP3 and 4

- From pilot data (unpublished and collected in December 2023), it was identified a clinically significant meaningful change in respiratory insufflation volume is 1.5mls/kg when adjusting MIE pressures. With a standard deviation of 3.2, we would need 36 patients to account for this difference as statistically significant with 80% power and 5% significance. To allow for 10% attrition (patient drop out, illness and death), we would aim to recruit 50 patients from all functional groups to ensure we recruit at least 36 patients. Our current population aligned with this project is 85 patients.

WP3: Feasibility and Impact of TFL for airway assessment and during MIE with children with SMA

WP3 will be divided into 2 stages:

a. Describe and evaluate upper airway response patterns with MIE.
b. Evaluate whether changing MIE settings can alter response patterns seen in the upper airway.

Sampling frame

- Patients known to GOSH based on the following inclusion criteria: Genetic confirmation of SMA, use of MIE for airway clearance for >3 months, care under respiratory physiotherapists at Great

Pre-submission

Ormond Street Hospital, including children of all age groups.
- Exclusion: Patients on palliative care pathways and/or patients who transitioned to adult services.

Methodology WP3a

A cross-sectional study.

Data collection will include (utilising existing databases):
- Age/Date of birth; Sex; Growth centile (Weight/Height); Copies SMN; SMA Treatment and age at treatment; Non-invasive ventilation (NIV) user/hours of ventilation; Chest physiotherapy techniques/nebulisers; Medications; Admissions and antibiotic use in the last six months; Compliance data to MIE /compliance to NIV; Existing secretion management plan/baseline MIE pressures and start date; Functional ability/motor function score; Scoliosis (Cobb angle) +/- surgery.

TFL assessment
- Participants will be invited by telephone and a follow up email.
- Patients will be monitored for oxygen saturations throughout the assessment.
- The laryngoscope will pass through a modified interface which attaches to the MIE, and then placed on the patients face.
- The laryngoscope, which is 2.8mm in diameter, will be supported and introduced to the patient trans-nasally.
- The examination will be visualised on a screen with continuously running video recordings from the TFL output and an external video camera will record the pressure screen of the MIE device, allowing for documentation of phases of insufflation and exsufflation (Appendix3).
- Video-recorded TFL will visualise laryngeal anatomy at rest (normal breathing) and during baseline individualised patient MIE settings (as determined prior to start of the study)
- Different levels of the upper airway (Glottic, Supraglottic and Hypopharyngeal) will be scored (0 abduction, 1 adduction) resulting the outcome of laryngeal responses to MIE.
- During the exsufflation, the cough peak flow will be recorded through a pneumotachograph.
- Other parts of equipment set up: a video camera monitoring of the patient and, a pneumotachograph and pressure transducer in the MIE circuit to give ongoing flow- and pressure tracing.
- Inter-rater reliability will be measured through review of recordings by a blinded assessor.

Timeline-Months 14-21

Data collection and analysis for WP3a:

Data collection will be targeted over 3 months, with the goal of testing 16 patients per month, maximum 2 patients per day. Data collection takes approximately 1 hour per patient.

Data will be collected on an existing evaluation form from previous work(26) to allow for review of each level of the larynx and analyse if it is abducted or adducted. It allows for comments and documentation to indicate the stage of MIE cycle.

Anonymised videos will be uploaded to ChilliPharm, a secure online platform specialising in clinical trials, and videos obtained from TFL will be uploaded alongside the video monitoring the pressure on the MIE device. They will be synchronised by a verbal cue, screened, and analysed by the principal investigator. The online platform also allows for inter-rater reliability testing.

Data will be recorded in REDcap, coded and transferred to R for statistical analysis. Baseline characteristics will be analysed using means and standard deviations, or number and proportion as appropriate. Between group demographic analysis based on functional levels will be analysed using ANOVA, Kruskal Wallis tests or Chi Square test.

Airway status will be analysed as abducted or adducted at each level of the upper airway and compared by functional groups (non sitters, sitters, standers and walkers combined) and analysed utilising Chi squared or Fishers exact test. The lowest open level with be compared between functional groups using Mann Whitney tests.

Incomplete data will be excluded.

Methodology WP3b

An interventional pre and post study, examining effect of changing MIE settings on the upper airway responses. Patients will start on the pressures identified in WP3a and following a protocol-driven stepwise approach, created for the purpose of this study based on previous work(18), settings will be adjusted to obtain highest pressures with

Pre-submission

an open upper airway. This provides data to characterise the upper airway at different levels of positive and negative pressure.

- TFL assessment together with flow- and pressure tracings, analysis and measurement of inter-rater reliability are performed as in WP3a.
- Patient will start on baseline MIE settings. A protocol will be followed to obtain best pressures on MIE with an open upper airway. These settings will be adjusted depending on the view seen at baseline. If the airways are open at baseline, MIE device settings will be increased in 2cmH_2O increments until airway closure is noted. The settings will then be reduced to the last open level documented. If the airways are closed at baseline, MIE settings will be decreased in 2cmH_2O increments until airway opening is seen. These settings will be documented as the patients optimal settings.
- All patients will be asked to report the level of comfort of the different settings using an appropriate scale.

Sampling frame:

All patients from WP3a will be eligible to take part.

Exclusion:

- Non tolerance of TFL

Timeline-Months 14-21

Data Collection and analysis:

Data collection will occur simultaneously as WP3a and will occur at a range of settings on the MIE device, during one session unless patients require rest periods.

Airway status will be analysed as open (abduction) or closed (adduction) at each level of the upper airway at the different settings with clear documentation of the maximal settings. The maximal setting between groups will be analysed and compared utilising a T-test or Mann-Whitney test.

Cough peak flow from baseline settings and maximal modified settings will be compared and analysed with paired t-test or Wilcoxon matched pairs test.

Data will be analysed for relationships between age and SMA treatment plans.

WP4: Reviewing lung function of patients with OEP at baseline and adapted MIE settings and gathering information about acceptability and tolerance to TFL/OEP.

Methodology 4a

The child will be positioned in supine on a plinth with their thorax and abdomen free of clothing and temporary markers will be placed in specific locations, according to anatomical landmarks and existing protocols, throughout the thorax and abdomen. They will be distracted to remain relaxed.

The temporary markers are picked up by infra red cameras and signals are translated through computer software to recreate the shape and movement of the chest wall.

Data will be obtained whilst the child is completing:

- Spontaneous quiet breathing for 5-minutes
- A spontaneous cough
- Baseline settings of MIE
- Repeated quiet breathing for 5-minutes
- Newly adapted MIE settings
- Further 5-minute period of quiet breathing
- Respiratory function data: Tidal Volume(TV), Respiratory Rate(RR), Forced Vital Capacity(FVC), Forced Expiratory Volume(FEV1), Total lung capacity(TLC), compartmental chest wall volumes and Cough peak flow(CPF) will be collected during all tests

The testing should take 30 to 45 minutes, within an allotted 1.5 hours assessment time.

Sampling frame:

A follow-on plan from WP3, completed on the same day after a break, as suggested by PPIE work. All patients from WP3 will be eligible.

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Exclusion:
- Non tolerance of TFL from WP3
- Unable to complete period of quiet breathing.

Data Collection and Analysis

Data is recorded through the computer programme EVO. It will be analysed using a programme called SMART TRACKER.

The expectation with optimised MIE settings will improve respiratory function data.

Respiratory function data from original MIEs settings will be compared to adapted settings and will be analysed using a paired t test or a Wilcoxon matched pairs test.

Methodology 4b
- After a break, patients and families will attend a 30-minute recorded semi-structured interview regarding acceptability of TFL and OEP.
- If they did not tolerate the TFL, the semi-structured interview will occur following that section of WP3.
- A topic guide will be developed using themes from WP2, existing literature and discussion with the steering group including PPIE members and supervisory team.
- Interviewers will introduce the topic and encourage discussion, give prompts and clarification. Participants will discuss each point and feedback will be compiled into key areas.
- Utilising a range of approaches, views of the family and child will be captured. For children, this includes drawing/writing, playing, picture cards/talking mats and emoji scales, recognising younger children may not feel confident in joining open ended question discussions or answering direct questions.

Sampling frame:

A follow-on plan from WP3, completed on the same day after a break, as suggested by PPIE work.

All patients from WP3 will be eligible.

Exclusion:
- No patient will be excluded.

Data Collection and Analysis

Data collected from semi-structured interviews will be transcribed and inputted into NVIVO software to store and organise data. Inductive thematic analysis will be used to analyse data collected from the semi structure interviews. Data collection and analysis will occur concurrently through an iterative process enabling data already collected to inform subsequent data collection. This constant comparative methodology will extend to thematic analysis of each data set until information power is achieved(31).

Data from focus groups in WP2 will be anaylsed. This gives themes from family reported experiences of respiratory management and ideas regarding information clinicians should obtain. Data from the semi structured interviews in WP4b will be synthesized to give themes to develop appropriate questions to gather information regarding clinical tests regarding respiratory function. These will be combined to identify key themes relevant to clinical needs and population wellbeing.

Timeline-Months 14-21

Output, anticipated impacts and dissemination

Output

Improved clinical decisions.

A constant challenge for clinicians is set up of MIE devices and pressures required for optimal airway clearance. Utilising TFL will give us information regarding the upper airway of these patients and a range of parameters for MIE settings. For the first time in children with SMA, we will have direct measures of lung volumes utilising OEP.

Collecting key themes relevant to respiratory needs and population wellbeing will enable future steps towards creating a validated respiratory questionnaire for SMA to better understand the impact of respiratory management

Pre-submission

on these patients', families' and carers'.

Future study designs

Enabling access to patients across the United Kingdom, a comparison study between SMA patients who follow standards of care compared with MIE adjusted under direct visualisation with TFL, utilising OEP to highlight between groups results.

Work to translate the themes from the qualitative research into a validated respiratory questionnaire, it would be tested for sensitivity and validity.

Anticipated Impact

Personalised settings for MIE would reduce risk of dynamic airway collapse during treatments. It allows future prescribing of MIE to be more specific, improving secretion clearance, reducing risk of infections, and reducing dependence and associated costs on NHS services. In turn it will improve the health and wellbeing of the child. It reduces potential time spent in hospital and the associated disruption to patients, parents and family members, including education and work commitments. Developing a complex airway clinic will improve patient care and generate income for the trust.

Potential translation to other conditions requiring MIE, helping more patients. This project provides evidence on safety and efficacy of MIE in the paediatric population. It enables the professional community to agree consensus on MIE setting prescriptions where there is a gap in understanding, impacting on future policies and guidelines.

Dissemination

Parents/Patients/Charity dissemination

Parents and patients highlighted importance of sharing feedback. They have provided feedback for study design and will continue involvement within the steering group.

They provided insight into preferred dissemination methods. We will ensure allowances are made for those people whom English is not their first language. Some methods are highlighted below:

- Lay summary for parents/carers and patients in a leaflet/infographic format via coffee mornings, notice boards and parent conferences
- Online reports for relevant charities and GOSH websites
- Podcasts/Talking heads presentations
- Participation at GOSH/BRC research events
- Translation into a variety of languages

SMA UK and MD UK will support dissemination by sharing findings on websites and social media, engaging families from a diverse range of backgrounds.

Conference presentations and publications

Conferences will be targeted to reach families and multi-disciplinary audiences and include:

- International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress
- World Muscle Society
- SMA Europe
- CureSMA
- European Respiratory Society

A minimum of four journal articles are planned. Targeted journals include; Thorax, American Journal of Respiratory and critical care medicine, and Neuromuscular disorders.

International influence

International charities, advocacy groups and networks exist for SMA, and links with these ensure international output of study results.

This ensures information and results will be appropriately fed back and are widely accessible to all target audiences, maximising impact.

Ethical Considerations

Pre-submission

Full NHS ethical approval will be sort for WP2, 3, and 4 in months 1-7. The supervisory team will be fully involved ensuring ethical approval is secured in a timely manner. The PPIE steering group will be asked to review any documentation associated with the consent process.

Written informed consent will be obtained from parents/patients. Parents of children under 16yrs will be asked for written consent, young people over 16yrs will have the opportunity to provide written consent and children under the age of 16yrs will be offered the opportunity to assent. Participant information sheets will be designed to reflect age and developmental stage of children and young people. Easy read versions will be avaliable.

TFL is an invasive procedure with minimal risk factors. We will utilise videos and pictorial diagrams to ensure understanding of procedures.

OEP is non-invasive and requires children to stay relaxed. Pictorial diagrams will be used to explain the procedure and ensure understanding.

As many children are existing patients in the hospital, to avoid coercion, the patient and family will be invited to take part in the study by a independent clinician during a routine visit.

Maintaining Anonymity and Confidentiality

Data protection laws will always be adhered to. Confidentiality during each WP will be assured, including participants are not identifiable during dissemination. Data will be anonymised to avoid participant identification. An encrypted digital recorder will be used for interviews.

Data Protection and Security

Data will be handled in accordance with the Data Protection Act 1998 and trust policy. Data will be stored securely using a password protected computer and encrypted data stick/hard drive only the research team will access.

Success Criteria

WP1:

Objectives:

- Complete a comprehensive systematic review of the literature focusing on different aspects of this project in order to inform study design and development.

No risk identified.

WP2:

Objectives:

- Understand patients, parents, and carers experiences of day-to-day respiratory management including use of their cough assist machine
- Further understand views of children/families on important aspects of respiratory symptoms and well-being

Risks:

- Lack of recruitment of a diverse population
- Insufficient numbers of recruitment
- Potential for prior relationships with participants

Mitigation of risks:

- Ensure each category of SMA is represented
- Ensure participants represent diversity of the population
- Utilise associated networks to encourage participation and recruitment
- Engagement in self-reflexivity
- Presenting self objectively and neutrally

WP3 A and B:

Objectives:

- Laryngeal response patterns to MIE in SMA
- Physiological understanding of airway status of SMA children at rest and on utilising MIE
- Maximal airway pressure from MIE with all levels of airway open compared between functional groups

Pre-submission

Risks:
- Recruitment/retention of participants
- Anxiety related to unknown tests

Mitigation of risks:
- Timely consent with adequate resources i.e. videos to explain processes
- Equipment available at time of consent
- Appropriate time for assessments

WP4:

Objectives:
- 4a) Assess changes in respiratory function with optimised MIE settings.
- 4b) Gain information related to the new assessments to enhance the respiratory feedback from WP2.

Risks:
- 4a)Poor tolerance to lying flat.
- 4a)Lack of understanding to not remove markers.
- 4a)Lack of understanding of requirements during tests.
- 4b)Recall bias
- 4b)Fatigue from day of testing

Mitigation of risks:
- 4a)Equipment allowing for comfort.
- 4a)Suitable distraction techniques
- 4a)Parental support during tests
- 4b)Semi structured interviews completed the same day as testing
- 4b)If required, the interview may take place virtually the following day

Pre-submission

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

This proposal was developed in collaboration with parents, carers and patients of Great Ormond Street and charity partners from across the United Kingdom who are linked to SMA. Information has been shared regarding project proposal, study design, concerns and project relevance.

Six parents, three patients and two charity partners consulted on the plain English summary and joined focused discussions.

The Child and Young Person Group for Great Ormond Street Hospital contributed to the preparation of the application by advising on the plain English summary.

The families involved confirmed importance of the research and their feedback has informed development and design of the study from the outset.

- Families explained it is research that would be advantageous in helping their children stay out of hospital.
- One family reported it 'gives us a door to open up and see what is happening inside, it would be the best.'
- Another commented 'we are grateful you are helping and looking into the respiratory side of SMA'.

Below are some concerns raised and revisions/additions made as a result:

- Some patients and parents expressed concern regarding the transnasal fibreoptic laryngoscope. The concerns were regarding appearance, feel and what happens if the child dislikes it. A child friendly video/cartoon to explain the procedure step by step was suggested with a QR link so it can watch in clinic or at home. At the consent visit, patients requested to see the equipment that will be used. It will be accessible to all patients.
- Some families were concerned over their child's fatigue with multiple visits. They requested testing for the transnasal fibreoptic laryngoscope be completed in one visit with breaks, followed by the opto-electronic plethysmography in the afternoon. They expressed all testing and semi structured interviews be completed in one day with no other booked appointments.
- The parents, families and charities requested they receive regular feedback on the study, ensuring it was appropriate and all-inclusive.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Ongoing PPIE

Undertaking/Managing the Research

A steering group will be created comprising of 10 participants, made up of a combination of approximately 3 ex-patients (patients who have transitioned to adult services), 4 parents of children with SMA and 3 representatives from charity partners. All parents involved in the development stage have expressed an interest in joining the group. One charity partner has also expressed an interest. Further recruitment will be through email correspondence and public engagement on social media. Anyone interested will be offered a follow up meeting.

Parents/Patients/Charity Partners will have the opportunity to:

- Partake in training for being involved in an advisory group
- Guide the project throughout the research process
- Explore ethical issues
- Assist with plain English summaries/updates
- Provide patient representation, parent perspective and SMA network links.

This group will meet 3 times face to face (once a year) and online groups approximately 5-6 monthly and will be consulted via emails/telephone calls at other key points during the project.

Dissemination

Where appropriate the steering group will be invited to assist with writing final reports/summaries of the results and will be acknowledge for their involvement in the study. Their role in facilitating publication of the findings on charity/SMA specific websites and dissemination through their informal networks will be invaluable and key in ensuring appropriate dissemination to all audiences.

Pre-submission

Training and Support

The applicant will attend the PPIE training within the first six months of the fellowship to increase knowledge of PPI and its practical application. Training needs will be identified and supported by the applicant's current networks (GOSH PPI Lead/academic supervisors).

When recruiting parents/carers to the steering group a clear description of the roles/responsibilities will be provided, plus the opportunity for discussion. They will be offered training and support for their role in the steering group. PPI expenses will be calculated at the recommended NIHR rate.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

Pre-submission

8. Training & Development and Research Support

Proposed training and development programme

The focus of this plan is to ensure my academic and clinical needs are met, enabling progression as a clinical academic leader within neuromuscular respiratory physiotherapy. A comprehensive strategy has been developed including formal PhD academic tuition, specialist training, conferences, and an overseas visit. The training plan will enable me to complete robust and enhanced doctoral study and is integrated in the Gantt chart.

Academic Training

University College London (UCL) will host the PhD and has a world leading reputation and registration on the graduate research development programme will provide access to an extensive range of online resources and taught courses. My training and development plan has been created using the VITAE Researcher Development Framework, which ensures my research needs are met within all four domains. These encompass knowledge and intellectual abilities, personal effectiveness, research organisation and governance, and communication and impact. The formal and additional academic training, outlined below, has been developed to cover all aspects of work package one to four.

Formal Academic Training - part of the UCL PhD programme

Year 1
- An introduction to R with RStudio
- In a Nutshell: starting an NVivo project
- Introduction to REDCap for research
- Data import and export in REDCap
- Managing Participants in REDCap
- Getting started with your systematic review
- How to do PPI: focus groups and more
- The Writing Series: Academic Writing logic
- Introduction to Research Integrity and Support
- Mixed methods approach in social sciences

Year 2
- Introduction to qualitative analysis: Interviewing
- Introduction to qualitative research method: thematic analysis
- Engagement and Impact: Inclusive practice
- Introduction to writing your PhD thesis.

Year 3
- Academic Career Planning for PhDs
- Networking Skills
- Researchers skills beyond academia: leadership

Additional Academic Training

These additional opportunities will allow upskilling in specific areas relevant to my project.

Year 1
- Mixed methods systematic review training, University of Leeds, faculty of medicine and health
  - Focused on the mixed methods systematic review.
  - This course will provide me with in depth understanding of all aspects of completing a mixed methods systematic review, helping shape not only this research project, but future studies as well.
- Statistics courses: Chi-square and Beyond for 2x2 Tables at UCL
  - This course is designed for Healthcare professionals who already have some understanding of basic statistical principles and wish to develop their statistical expertise.
  - I will learn specific in depth analysis regarding chi-squared analysis which are statistical methods vital for this study and studies in the future.
- PPIE training workshops, UCLH
  - This course has a variety of workshops which enables a wide understanding of PPIE
  - It will ensure I link all aspects of the project to PPIE, allowing for feedback and adaptations as required, supporting the development this and future projects.

Pre-submission

    - It will also help me ensure I get the best results from the PPIE undertaken, including that of the steering group and other PPIE activities.
- Introduction to Mixed Methods Research, followed by Advanced Mixed Methods, Methodical via NCRM/UCL.
    - This course has an introductory level following into an advanced level. It is recommended by UCL and will give me in-depth knowledge of all aspects of my mixed-methods study, I plan to attend this module early in the fellowship. The module focuses on how to design, implement and analyse a mixed methods study to answer the individual clinical question.

Year 2

- Qualitative Research Methods, University of Oxford
    - Focused time on qualitative research methods.
    - It is essential for data collection with children and young people in WP2 and WP4b.
    - It will provide me with practical ideas for data collection, data analysis and ethical considerations when using qualitative research methods.

Year 3

- Development of leadership skills
    - Leadership Course: Emerging Clinical Leaders, The Kings Fund, 6 days
        - This course is renowned for supporting clinical leaders.
        - It will provide me with strategies to help lead future clinical academics.

Clinical Training

Throughout the fellowship in addition to my academic training, I will also advancing my clinical knowledge and skills.

Planned training is below:

Year 1

- Development of advanced clinical skills
    - Beyond Basic Airway Course and Bronchoscopy course
        - These course are renowned for teaching enhance airway skills which will be essential foundation blocks for WP3. These will then be further developed with Tiina Andersen.
    - Clinical Training for trans-nasal flexible laryngoscopy with Tiina Andersen
        - Tiina developed the testing procedure and will be key in teaching and developing my clinical skills. Two days would be spent observing, followed by six days of hands-on teaching and skill development, and two days of competence sign off
    - Clinical supervision with Colin Butler, Ear nose and Throat Surgeon at Great Ormond Street Hospital
    - Clinical time at NMCCC, Queens Square observing their Trans-nasal fibreoptic laryngoscopy clinic (2 days)
        - Purpose of visit
            - Observation of physiotherapy led Trans-nasal fibreoptic laryngoscopy in practice
            - Collaboration with physiotherapists using Trans-nasal fibreoptic laryngoscopy
            - Engagement and practical ideas of how to integrate Trans-nasal fibreoptic laryngoscopy within own clinical practice

Year 2

- Reciprocal visit from Dr Andersen for supervision in own paediatric setting
- 1-2 days per month carrying complex caseload

Year 3

- Maintenance of advanced clinical skills and competence
    - On-going clinical supervision with Colin Butler (Joint clinical sessions, case study reviews and ongoing clinical support

Conferences

- International Conference on Home Mechanical Ventilation and European Respiratory Care Association Congress JIVDERCA 2025
    - This International multidisciplinary (MDT) conference will offer clinical learning and networking.
    - The conference only meets every third year
- CureSMA 2025
    - This international conference is a family centre conference and would be an excellent opportunity to discuss the study to continue PPIE
- WMS 2026
    - This International conference is an excellent platform to disseminate study findings to date
- ERS 2027
    - This conference will be an excellent opportunity to disseminate study findings in the final year of

Pre-submission

the PhD.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Giovanni Baranello

Institution	Institute of Child Health, UCL

Position	Professor of Paediatric Neuromuscular Disorders

Description of support provided
As a Professor of Paediatric Neuromuscular Disorders, I am delighted to support Lisa Edel's PhD application. As a clinician academic, I am involved in the diagnosis and clinical management of children with different neuromuscular conditions at the Great Ormond Street Hospital. I am principal investigator in numerous neuromuscular clinical trials, predominantly in the areas of Spinal Muscular Atrophy and Duchenne Muscular Dystrophy. My main area of research relates to translational research in neuromuscular conditions, development of outcome measures and biomarkers, natural history studies and long-term studies in patients receiving novel treatments. With my clinical and scientific/academic background I am well positioned to provide Lisa with the right level of supervision during a PhD. I have supervised six PhD student, one to completion. I work closely with Lisa and we have collaborated on several projects, specifically the correlation between a clinical respiratory score and more objective respiratory assessments. Lisa has a strong interest in developing research in the respiratory management of children with neuromuscular diseases and has brilliant ideas on how to demonstrate respiratory physiotherapy management of airway clearance can lead to improved respiratory status of children with Spinal Muscular Atrophy. Lisa is intellectually excellent, hardworking, well-organised, with a keen and enquiring mind. She has excellent written and oral presentation skills and interacts well with colleagues. I recommend her unreservedly for this PhD application. She is outstanding in terms of her personal and intellectual attributes. She will be an excellent PhD student and will progress to a successful clinical-academic career.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Tiina Andersen

Institution	Western Norway University of Applied Sciences, Bergen, Norway

Position	Associate Professor

Description of support provided
Tiina is a clinical specialist in cardiopulmonary physiotherapy with 25 years clinical experience from specialist

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healthcare, including 20 years' service at the Norwegian National Advisory Unit on Home Mechanical Ventilation at Haukeland University Hospital, Bergen, Norway. She has worked as associate professor since 2019.

Her clinical, teaching, and research interests are in respiratory physiotherapy, where she has worked extensively in acute and chronic care. She is one of the world's leading physiotherapists in chronic respiratory failure, including being frequent invited speaker at international forums, and serving as the physiotherapy representative in organizing committee of World Congress on Home Mechanical Ventilation (JIVD Congress), as well as the 'Respiratory Failure and Mechanical Ventilation' course for the world's largest lung organization, European Respiratory Society (ERS).

Tiina has worked extensively with transnasal fibreoptic laryngoscopy (TFL) and mechanically assisted cough device examining laryngeal responses, establishing norms in healthy individuals and identifying abnormalities in adults with ALS (Amyotrophic Lateral Sclerosis). She has worked as a postdoctoral researcher, continuing investigating respiratory support improvements for ALS with an international multicentre study. At this timepoint, Tiina supervises 5 PhD students and she is keen to serve as an academic supervisor to Lisa Edel to further implement upper airway visualization in a large paediatric SMA cohort. Tiina will attend monthly supervisory meetings and will be responsible for the training of TFL. Tiina's international experience will provide a broader perspective within the supervisory team.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Colin Butler

Institution	Great Ormond Street Hospital

Position	Ear Nose and Throat Surgeon

Description of support provided
I am delighted to support this research proposal which has the potential to significantly advance the care of children with Spinal Muscular Atrophy (SMA). The research proposal has the potential for high impact in improving respiratory health not only in children with SMA but has application to other pathologies. Lisa is in unique position at Great Ormond Street to investigate children with SMA and has an excellent track record for research to date. I am extremely happy to offer assistance in supervising the clinical application of endoscopy techniques of the proposed work, and will assist Lisa with the analysis and research methodologies proposed. Lisa will have access to the Ear Nose and Throat (ENT) research group at ICH, including weekly research group meetings with airway-based research clinicians. I co-lead the EpiCENTR research group at ICH which focuses on paediatric ENT research, where we have an active group of PhD and MSc students. I currently co-supervise 2 PhD students and 2 MSc students. If awarded, Lisa will have regular supervision and gain peer support from the doctoral students linked to our group. A benefit of the cross-discipline approach will mean she will have support from several allied specialties. As an ENT consultant focusing on airway research, I can provide clinical training in endoscopy techniques Lisa requires for the proposal. I am enthusiastic in offering support to a project I believe to be capable of improving healthcare, not only for the targeted area, but in general for children with significant respiratory disease.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Paula Kelly

Institution	Great Ormond Street Hospital

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Position	Research Associate, ORCHID

Description of support provided
Dr Paula Kelly is a Clinical Academic at Great Ormond Street Hospital (GOSH), she is a children's nurse and a senior research fellow within the Centre for Outcomes and Experience Research in Children's Heath, Illness and Disability (ORCHID) at GOSH. She is an experienced qualitative researcher, focusing on the heath care of children, young people and their families experiencing serious illness. She will bring additional expertise in undertaking qualitative research with families to Lisa's team, also providing ongoing support to planned PPIE work in the application. Dr Kelly has held academic posts as a lecturer in child health nursing at Kings College London including experience of supervision of undergraduate and post graduate research dissertations. She is co-author of 23 peer review journal papers. As a senior research associate in The Louis Dundas Centre for Children's Palliative Care at the Institute of Child Health University College London, she managed several research studies including those with a mixed methods design. Dr Kelly's commitment and experience in supporting nurses and allied health care professionals to develop clinical academic careers is evident from her association with the NIHR BRC at GOSH and ORCHID since 2017 providing programmes of activities and structured support to enable clinical academics to fulfil their research potential. She has worked with Lisa as a clinician and provided support to develop aspects of this application. As an experienced clinical academic she will provide regular monthly practice supervision and opportunities for additional contact as required.

PRACTICE SUPERVISOR 2

Name	Dr Martin Samuels

Institution	Great Ormond Street Hospital

Position	Respiratory Consultant

Description of support provided
Dr Samuels is a Consultant Respiratory Paediatrician at Staffordshire Children's Hospital in Stoke-on-Trent and at Great Ormond Street Hospital in London. He has been in practice for over 40 years. He interests are in respiratory disorders in children, including sleep disordered breathing, neuromuscular and congenital central hypoventilation syndrome (CCHS). He has undertaken many research studies and has over 75 peer-reviewed publications. He worked in PICU for over 20 years and was Chair of the APLS International Working Group for 11 years. He has contributed to guidelines on Respiratory Control Disorders (RCPCH) and Sleep Disordered Breathing (BTS). He has overseen the development of sleep services in GOSH and Staffordshire over the last 10 years, and recently completed GOSH-funded research on the effects of flight-induced hypoxia in CCHS and neuromuscular disorders. He is currently developing the use of phrenic nerve pacing for central hypoventilation disorders as a national service in the UK. He lectures regularly on Sleep & Respiratory Control Disorders, as well as Long Term Ventilation nationally and internationally. Dr Samuels is keen to support as a practice supervisor as he is able to bring his experience as a clinical academic in the care of children with respiratory compromise. He will provide additional expertise in the analysis of lung function and clinical outcomes. He will meet on a monthly basis and actively participate in supporting Lisa's overall development as an early careers researcher.

PRACTICE SUPERVISOR 3

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Name	Mrs Ema Swingwood

Institution	University Hospitals Bristol and Weston, University of West of England

Position	Consultant Physiotherapist and Clinical Research Fellow

Description of support provided

Ema is a consultant therapist for critical care at University Hospitals Bristol and Weston. Her areas of clinical and teaching expertise include ventilation, weaning and complex airway clearance. She has 20 years of clinical experience specifically in respiratory (both adults and paediatrics) and has been extensively involved in cough augmentation and airway clearance using the mechanical insufflation-exsufflation device, so has a good understanding of the clinical area that Lisa wishes to explore. She has lectured at International conferences over a number of years around these topics. Having completed the Advanced Cardiorespiratory Physiotherapy MSc programme at University College London (UCL) in 2012, Ema has continued her research focusing on the use of Mechanical Insufflation-Exsufflation (MI-E) and other cough augmentation strategies. Her PhD work explored the use of MI-E in the intubated population, which was funded through a NIHR Clinical Doctoral Research Fellowship. This is due for completion in July 2024 (thesis already submitted).

Ema is also the research lead for Diagnostics and Therapies where a key component of her role is embedding research into the clinical setting and developing a research infrastructure for non-medics. Ema has supervised MSc students (University College London) to completion (8) and is named on current ICA Internship (1) and Pre-Doctoral applications (1) as a clinical academic supervisor.

Her experiences to date will enable her to work effectively with Lisa during and beyond her PhD to develop a clinical academic role.

Collaborations

The national and industry collaborations I intend to establish are outlined below. These will support and enhance my research project, and training and development programme, whilst also ensuring I develop partnerships for future research.

Haukeland University Hospital, Bergen, Norway

Haukeland University Hospital is the second largest university hospital in Norway, which functions as a full service public teaching hospital. The hospital provides significant training and research activities housing several specialty units that are internationally recognised as centres of excellence. Haukeland University Hospital endeavours to translate cutting edge research and teaching into improved patient outcomes and the best quality clinical care. A visit has been scheduled for the first year of my project and I am fortunate to Tiina Andersen as an academic supervisor who is based at Haukeland University Hospital. The visit will focus on training and collaboration to develop ongoing research development. The visit will focus on learning the technique for the project and working in the laboratory. It will reinforce links with European collaborations.

Professor Paul S McNamara, Professor of Child Health and Consultant in Paediatric Respiratory Medicine based at University of Liverpool and Alder Hey Children's NHS Foundation Trust Hospital

Throughout the course of the fellowship, a strong working relationship with Professor McNamara will be developed. He is an experienced clinical academic who has specialised in the development of care questionnaires. He provides support and knowledge regarding development of topic guides and qualitative methodology. Linking with the largest children's hospital in the United Kingdom will provide more opportunities to work closely with the aim to contribute to research learning programmes during the fellowship and in the future.

Collaborations from Industry

This project will link with different industry partners. This enables future potential for new opportunities, resources

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and building a network beyond the public sector. It will enhance the impact and relevance of the proposal.

ChilliPharm This company will be providing the online video platform for video synchronisation. They are already well know in the medical field and have collaborated with allied health professionals in the past. They are leaders in utilising specialist videos for clinical trials and are happy to be collaborators on this proposal.

Breas This company design and produce the mechanical insufflation devices. They are the leading company in the United Kingdom. They have worked alongside Great Ormond Street Hospital to provide these devices to patients and are happy to continue to support this project. They are an international company and have excellent networking opportunities.

BTS Bioengineering The opto-electronic plethysmography equipment was provided by BTS bioengineering and they are willing to continue to support the equipment throughout this project.

Storz The trans-nasal fibreoptic laryngoscopes are managed by Storz. They provide excellent on site support at Great Ormond Street and are happy to collaborate on this project.

Host Organisation Support Statement
The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

The partnership between the Physiotherapy Department at Great Ormond Street Hospital (GOSH) and the Developmental Neurosciences research department at the Institute of Child Health (ICH), University College London (UCL) will provide an outstanding environment for Lisa to complete a doctoral fellowship and build a strong clinical academic career. In addition physiotherapists contribute significantly to a range of UCL MSc programmes, providing specialist post-graduate lectures, offering clinical placements, and co-supervision of MSc dissertation projects. We confirm that, if the award is granted, Lisa will be integrated in both the Physiotherapy service at GOSH and the ICH enabling her to conduct clinical and research development activity appropriate to the fellowship.

Our vision at GOSH is to become a research hospital. It forms an integral part of the working lives of our staff and the patients/families we treat. We learn from each patient, using the knowledge gained to improve the health of patients at GOSH and children across the world. Research is considered a core component when recruiting to leadership positions across the organisation. All clinical divisions are supported to develop and own their research agenda and the Physiotherapy service is no exception; pro-active in both leading and collaborating on research, and development with other members of the multidisciplinary team at GOSH, and national and international centres.

GOSH has on-going physiotherapy research projects (many with external funding) within all specialities. The Physiotherapy department supports staff to attend and present at national and international events, ensuring dissemination of research activity. We have extensive experience of supporting staff in pursuing a clinical academic career through the NIHR fellowships (3 CRDF's, 2 PCAF'S and 1 DCAF). As a result, we have built a questioning workforce with a commitment to drive change and improvement within and outside of GOSH for the benefit of patients, parents, staff and the wider NHS.

We have had 5 doctoral prepared physiotherapists who continue to pursue clinical academic and research career's and a further physiotherapist in the first year of a of a NIHR Doctoral Clinical and Practitioner Academic Fellowship. A robust cohort to provide excellent peer support to Lisa during her fellowship and in her future clinical academic career.

GOSH Physiotherapy is fully supportive of Lisa's application for this award; not only is this an excellent opportunity for her development but it also brings benefits to Physiotherapy and the Neuromuscular Respiratory Services at GOSH and the wider Trust. Lisa will continue to be embedded within the physiotherapy department contributing through regular clinical sessions, enabling her to maintain her high level of expertise in neuromuscular respiratory care, together with continuing to support and be a role model to colleagues involved in workplace research and audit activity.

As part of GOSH, Lisa will be able to access support from a variety of other sources. The Division of Research and

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Innovation sits alongside the Trust's clinical divisions and provides a one-stop shop for researchers, along with infrastructure to support research teams both in developing and delivering research projects. Lisa will continue to obtain support from the Centre for Outcomes and Experience Research in Children's Health, Illness and Disability at GOSH. This dedicated research unit is led by Professor Faith Gibson, Deputy Chief Nurse for Research, who holds a joint appointment with the University of Surrey. The team comprises of Senior Research Fellows from differing allied health backgrounds and focuses on delivering research that seeks to improve the care, health and well-being of children and young people. Lisa would have access to the clinical academic careers programme which provides education and support specifically for nurses and allied health professionals. She will benefit from the experience of core members who have been Chief Investigator or co-applicants on NIHR/Wellcome/MRC funded projects, together with many other charity funded projects. Outputs from the Centre during this time include over peer-reviewed publications and conference presentations. They have also supported clinicians in securing funding at varying stages on the NIHR clinical academic pathway and would provide Lisa with on-going support beyond the NIHR BRC fellowship,

The Great Ormond Street Institute of Child Health (ICH), which is part of the Faculty of Population Health Sciences within the School of Life and Medical Sciences at UCL is exceptionally well equipped, both in terms of facilities and expertise to support this fellowship. The ICH together with its clinical partner GOSH, forms the largest concentration of children's health research in Europe. As a PhD student within the developmental neurosciences section Lisa would be integrated into a thriving academic community. UCL and ICH have excellent record of research with top class research facilities. In the Research Excellence Framework UCL was the top-rated university in the UK for research power (average research score multiplied by staff numbers) and also had the greatest 'impact power' of any university.

The department currently hosts PhD students and has a strong track record of supporting NIHR funded students. The facilities available at ICH/UCL and GOSH fully meet the range of requirements for hosting an NIHR Fellow.

Within UCL Lisa will have access to a network of 18 libraries with an excellent support network for research and academic writing. The section has vast experience of clinical physiotherapy and paediatric critical care research providing further teaching opportunities, interaction with fellow researchers working towards common goals, and guidance and mentoring from recognised physiotherapy, neuromuscular and respiratory experts. The question posed by this research is vital to on-going developments in respiratory management of children with spinal muscular atrophy and the patient's effectiveness of airway clearance and quality of life. Lisa has extensive clinical experience in this area and is in an optimal position to complete this research, and as such we provide strong support for this application. UCL encourages applicants to push frontiers; work with integrity; be sensitive to diversity; and engage with society beyond academia.

Supporting Lisa to undertake a fellowship hosted by GOSH and partnered by UCL will further strengthen the department's relationship, thus ensuring clinically relevant research is completed which has a direct impact on patient care and healthcare service provision.

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9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Lisa Edel

Employing organisation	Great Ormond Street Hospital For Children NHS Foundation Trust

Grade	7

Starting point on scale	50056.00

Increment Date	17/10/2024

Justification of Costs
Will be undertaking the fellowship full-time for 36 months hence the 1FTE funding request which is in line with the terms and conditions of the funding call.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	53621
Geographical Weighting (£)	7746
Other allowances (£)	
Superannuation (£)	14593
Nat. Insurance (£)	7475
Current Annual Costs (£)	**83435**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mrs Lisa Edel
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Will be undertaking the fellowship full-time for 36 months hence the 1FTE funding request which is in line with the T&Cs of the funding call.

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	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	53621	53621	53621	160863
National Insurance	7475	7475	7475	22425
Superannuation	14593	14593	14593	43779
Geographical weighting / Other allowances (if applicable)	7746	7746	7746	23238

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	0	0
Total NHS	250305	0	0	250305
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: International conference fees
Justification of Cost: Fees for international Conferences as outlined below; Cure SMA in 2025, WMS in 2026, ERS in 2027

	Year 1	Year 2	Year 3	Total Cost (£)
	200	200	200	600

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: Flights for overseas conferences
Justification of Cost: Overseas visits for face-to-face conferences

	Year 1	Year 2	Year 3	Total Cost (£)
	400	500	300	1200

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: Accommodation & Subsistence for overseas conferences
Justification of Cost: Cure SMA in 2025 = 3 nights 3. WMS in Japan in 2026 = 4 nights 4. ERS in 2027 = 4 nights

	Year 1	Year 2	Year 3	Total Cost (£)
	300	450	450	1200

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Flights for overseas visit
Justification of Cost: Overseas visit to Bergen, Norway. For face-to-face meeting, training and supervision with Dr Andersen (clinical supervisor) and to establish international networks - 8 day visit, Autumn 2025 following completion of systematic review Travel - £200 (average return price flights to Norway).

	Year 1	Year 2	Year 3	Total Cost (£)

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	200	0	0	200

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Accommodation & Subsistence - Overseas visit
Justification of Cost: Overseas visit Bergen, Norway. Accommodation - 8 nights @ £150/night = £1200 Subsistence - 8 days/nights @ £25 = £200

	Year 1	Year 2	Year 3	Total Cost (£)
	1400	0	0	1400

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: National Visit
Justification of Cost: 2 day visit to National Centre of Neurology and neurosurgery to observe the TFL clinic. Purpose of visit to observe physiotherapy led TFL and collaboration opportunities. Average daily London travelcard train ticket £15.20 x2 = £30.40.

	Year 1	Year 2	Year 3	Total Cost (£)
	31	0	0	31

Type of cost: NHS
Description: Journey Costs
Conference: False
Item Description: Travel to Oxford
Justification of Cost: Train costs for return to Oxford for qualitative statistics course train ticket £36.20 when bought on advanced saver

	Year 1	Year 2	Year 3	Total Cost (£)
	37	0	0	37

Type of cost: NHS
Description: Subsistence
Conference: False
Item Description: Accommodation & Subsistence - Oxford visit
Justification of Cost: Oxford Accommodation - 4 nights @ £150/night = £600 Subsistence - 4 days/nights @ £25 = £100

	Year 1	Year 2	Year 3	Total Cost (£)
	700	0	0	700

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	3268	1150	950	5368

Equipment

Type of cost: NHS
VAT relief to be applied: False
Description: Encrypted Voice recorder
Price excluding VAT £: 200
Expected life time: 36
Justification of Cost: For recording and playback of the focus meetings (Work Package 2 and interviews (Work Package 5)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	200	0	0	200

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Type of cost: NHS
VAT relief to be applied: False
Description: Headphones
Price excluding VAT £: 75
Expected life time: 36
Justification of Cost: For playback of the focus meetings (Work Package 2) and interviews (Work Package 5)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	75	0	0	75

Type of cost: NHS
VAT relief to be applied: False
Description: Encrypted hard drive and memory stick
Price excluding VAT £: 460
Expected life time: 36
Justification of Cost: Required for storage and back up of all project data. Cost is based on 2TB (£390) external hard drive and a 128GB memory stick (£70)

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	460	0	0	460

Type of cost: NHS
VAT relief to be applied: False
Description: Rental of equipment associated with transnasal flexible laryngoscopes
Price excluding VAT £: 28646
Expected life time: 36
Justification of Cost: Required equipment to complete the transnasal fibreoptic laryngoscopes described in work package 3 Leasing cost involves part purchase and part loan totally £28646 for a 4 month period VS. a purchase price would be £54350.40

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	0	28646	0	28646

Type of cost: NHS
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 1000
Expected life time: 36
Justification of Cost: This is for the lead applicant, for use in all aspects of project management and write up.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	1000	0	0	1000

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	1735	28646	0	30381

Consumables

Description: Cleaning of Transnasal fibre optic Laryngoscopes
Type of cost: NHS
Please describe fully: Cleaning of Transnasal fibre optic Laryngoscopes
Justification of Cost: Cleaning of equipment between patient use £38 per scope for 50 patients = £1900

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1900	0	1900

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Description: Double sided stickers for opto electronic plethysmography
Type of cost: NHS
Please describe fully: Double sided stickers for opto electronic plethysmography
Justification of Cost: Stickers for OEP Single use stickers for the OEP, 52 stickers required per patients per visit = 2,600 stickers required (£60 per 500, 6 packs required)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	360	0	360

Description: Poster Printing for conferences
Type of cost: NHS
Please describe fully: Poster Printing for conferences
Justification of Cost: Poster printing for abstracts submitted to conferences, based on 2 in year 2 and 2 in year 3 at £60+VAT = £72.00 each pre fabric poster

	Year 1	Year 2	Year 3	Total Cost (£)
	0	144	144	288

Description: Printing costs
Type of cost: NHS
Please describe fully: Printing costs
Justification of Cost: Printing costs for consent forms and patient information leaflets

	Year 1	Year 2	Year 3	Total Cost (£)
	100	100	100	300

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	100	2504	244	2848

Patient and Public Involvement

Description: Travel for meetings
Type of cost: NHS
Please describe fully: Travel for meetings
Justification of Cost: Steering group travel Cost for 10 members to attend 3 face to face meetings, average £75 per journey

	Year 1	Year 2	Year 3	Total Cost (£)
	750	750	750	2250

Description: Payment for time
Type of cost: NHS
Please describe fully: Payment for time
Justification of Cost: Steering group time: Cost for 10 members to attend 3 face to face meetings and a further 3 online meetings at £75 per half a day

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	1500	1500	4500

Description: Costing for steering group refreshments
Type of cost: NHS
Please describe fully: Costing for steering group refreshments
Justification of Cost: Refreshment for steering group meetings £20 per face to face meeting

	Year 1	Year 2	Year 3	Total Cost (£)
	20	20	20	60

Description: Costing for refreshments for patients and families during testing days
Type of cost: NHS
Please describe fully: Costing for refreshments for patients and families during testing days
Justification of Cost: Refreshment for patients and families for test days: £20 per family

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	Year 1	Year 2	Year 3	Total Cost (£)
	0	1000	0	1000

Description: Training for steering group
Type of cost: NHS
Please describe fully: Training for steering group
Justification of Cost: PPI – Steering group training Imperial Public Involvement Training – Public Involvement in Research | Coursera – Free for members of the public

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Talking Mats
Type of cost: NHS
Please describe fully: Talking Mats
Justification of Cost: Talking mats for age related feedback To enable all children to take part in consent and feedback - annual fee £120 for the complete package (https://www.talkingmats.com/new-digital-app-subscriptions/)

	Year 1	Year 2	Year 3	Total Cost (£)
	120	120	120	360

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	2390	3390	2390	8170

Other Direct Costs

Description: Dissemination - Podcast/Talking heads production
Type of cost: NHS
Please describe fully: Dissemination - Podcast/Talking heads production
Justification of Cost: Dissemination – Podcast/Talking heads production to ensure equitable dissemination of information

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	540	540

Description: Covidence Software
Type of cost: NHS
Please describe fully: Covidence Software
Justification of Cost: Access to Covidence software – up to 3 reviews and unlimited collaborators

	Year 1	Year 2	Year 3	Total Cost (£)
	503	503	503	1509

Description: NVIVO Software
Type of cost: NHS
Please describe fully: NVIVO Software
Justification of Cost: Access to NVIVO software for qualitative data analysis (£97 annual costs for students)

	Year 1	Year 2	Year 3	Total Cost (£)
	97	97	97	291

Description: Transcription Costs
Type of cost: NHS
Please describe fully: Transcription Costs
Justification of Cost: Transcription costs for qualitative data analysis (10 hours in year 1, 25 hours in year 2)

	Year 1	Year 2	Year 3	Total Cost (£)

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	1383	3456	0	4839

Description: Statistical Support
Type of cost: NHS
Please describe fully: Statistical Support
Justification of Cost: Statistical Support has been costed in to the study, based on a yearly rate: £900 - Year 1, £1800 - Year 2 and £2700 - Year 3

	Year 1	Year 2	Year 3	Total Cost (£)
	900	1800	2700	5400

Description: Video equipment and online Secure video platform and support
Type of cost: NHS
Please describe fully: Video equipment and online Secure video platform and support
Justification of Cost: 1 instance access to ChilliPharm, 1 iPad filming kit to be used for the external patient videos provided for free, support with synchronisation of videos from the room and the TFL to allow for data analysis (per video synchronisation £400) with set up costs of £2500. Free access to online platform with ongoing support

	Year 1	Year 2	Year 3	Total Cost (£)
	2500	20000	0	22500

Description: Creation of animation for study outline for participants
Type of cost: NHS
Please describe fully: Creation of animation for study outline for participants
Justification of Cost: Following on from PPIE, this cost is for the creation of a video for participants explaining the project (2 Minutes of digital 2D animation).

	Year 1	Year 2	Year 3	Total Cost (£)
	3000	0	0	3000

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	8383	25856	3840	38079

Training And Development

Type of cost: NHS
Description: Short Courses
Item Description: Mixed Method Trainins
Justification of Cost: Introduction and Advanced mixed method training (£200 per workshop)

	Year 1	Year 2	Year 3	Total Cost (£)
	400	0	0	400

Type of cost: NHS
Description: Short Courses
Item Description: PPIE training workshops
Justification of Cost: Free training for UCL staff

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Emerging clinical leaders course
Justification of Cost: Completion of this renowned leadership course, hosted by The Kings Fund, will provide a comprehensive opportunity for leadership skills development. The 2024 cost was £2195 +VAT = £2634.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2634	2634

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Type of cost: NHS
Description: Short Courses
Item Description: Mixed methods systematic review training, University of Leeds, faculty of medicine and health
Justification of Cost: Online course, course fees only

	Year 1	Year 2	Year 3	Total Cost (£)
	700	0	0	700

Type of cost: NHS
Description: Short Courses
Item Description: Qualitative Research Methods Course based in Oxford
Justification of Cost: Course fees

	Year 1	Year 2	Year 3	Total Cost (£)
	2250	0	0	2250

Type of cost: NHS
Description: Short Courses
Item Description: Statistics course for Quantitative research
Justification of Cost: Online course at UCL

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Type of cost: NHS
Description: Short Courses
Item Description: Bronchoscopy Course
Justification of Cost: 3 day course for learning to use bronchoscopies https://www.rbht.nhs.uk/events/school-bronchoscopy-three-days-0

	Year 1	Year 2	Year 3	Total Cost (£)
	1100	0	0	1100

Type of cost: NHS
Description: Short Courses
Item Description: Airway anatomy and management course
Justification of Cost: Beyond BASIC Airway management course to increase skills with management of the upper airway. Beyond BASIC Airway Management Course | Royal Brompton & Harefield hospitals (rbht.nhs.uk)

	Year 1	Year 2	Year 3	Total Cost (£)
	300	0	0	300

Type of cost: NHS
Description: Applicant PhD Tuition Fees
Item Description: Applicant PhD tuition Fees
Justification of Cost: The applicant will be undertaking the PhD full time at UCL and appropriate costs have been included.

	Year 1	Year 2	Year 3	Total Cost (£)
	6215	6400	6592	19207

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	10965	6400	9226	26591

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)

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Total Direct Costs	110276	151381	100085	361742

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	0	0	0
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	361742	0	361742
Total Research Costs Requested From Funder	361742	0	361742

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**83435**	**83435**	**83435**	**250305**
Travel-subsistence-and-conference-fees	**3268**	**1150**	**950**	**5368**
Equipment	**1735**	**28646**	**0**	**30381**
Consumables	**100**	**2504**	**244**	**2848**
Patient-and-Public-Involvement	**2390**	**3390**	**2390**	**8170**
Other-Direct-Costs	**8383**	**25856**	**3840**	**38079**
Training And Development	**10965**	**6400**	**9226**	**26591**
HEI-Indirect-costs	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**

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Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**110276**	**151381**	**100085**	**361742**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Excess Treatment Costs	0	33950	0	33950.0

Justification:
Excess Treatment Costs: £679.00 per participant (for ward fees as per SOECAT), with a forecasted overall total of £33,950 for 50 participants.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	0	33950	0	33950

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	-33950	0	-33950

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	-33950
TOTAL NHS SUPPORT AND TREATMENT COSTS	-33950

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	361742

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Total NHS (Support and Excess Treatment) Costs	-33950
Total Funding Required	327792

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10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - Reference Lists .pdf) is included as an appendix within this file.

(Figures / Tables - Overall Schematic of Study (Appendix 2)(3).pdf) is included as an appendix within this file.

(Research Timetable - Gantt Chart(3).pdf) is included as an appendix within this file.

(SoECAT Template - 23SH29 - IRAS 34988_SoECAT_TFL for SMA (CPMS 34988) (VALIDATED 24May2024).xlsx) is included as an appendix within this file.

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11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

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Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Giovanni
Surname	Baranello
Position	

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Tiina
Surname	Andersen
Position	Associate professor

Supervisor Participant 2

Title	Dr
Forename(s)	Colin
Surname	Butler
Position	ENT Consultant

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	Paula
Surname	Kelly
Position	Clinical Nurse Specialist

Practice Support Participant 2

Title	Dr
Forename(s)	Martin
Surname	Samuels
Position	Consultant Respiratory Paediatrician

Practice Support Participant 3

Title	Mrs
Forename(s)	Emma
Surname	Swingwood
Position	Respiratory Pathway Lead

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Miss
Forename(s)	Wing-Yun
Surname	Ng
Position	Finance Manager

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the

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required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Torsten
Surname	Baldeweg
Position	Professor of Cognitive Neuroscience

Head of Department or Senior Manager 2

Title	Dr
Forename(s)	Tim
Surname	Liversedge
Position	Chief of service

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12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
I have no conflicts or potential conflicts of interest

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

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Appendices

1. Reference Lists .pdf
2. Overall Schematic of Study (Appendix 2)(3).pdf
3. Gantt Chart(3).pdf
4. 23SH29 - IRAS 34988_SoECAT_TFL for SMA (CPMS 34988) (VALIDATED 24May2024).xlsx

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Appendix #1: Reference Lists .pdf

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Appendix #2: Overall Schematic of Study (Appendix 2)(3).pdf

Pre-submission

Year 1

Year 2

Year 3

Work Package 1

Months 1-6

Systematic Review

Months 1-6

- Study Set Up
- Governance & Ethical Approvals
- Steering Group Formation

Work Package 2 - Patient and Parent perspective of respiratory management

Month 7: Recruitment

Months 8-9: Data Collection

N = 24-30

Exploring experiences of respiratory management from patients and parents

- Focus Groups
- Data Collection
- Data Analysis

Work Package 3 Feasibility and Impact of TFL for airway assessment and during MIE with children with SMA

Months 14-16: Recruitment

Months 16-18: Data Collection

N = 50

Part A
Describe and evaluate upper airway responses with MIE using TFL.

Part B
Evaluate whether changing MIE settings can alter the response patterns seen in the upper airway.

Work Package 4A - Reviewing lung function of patients with OEP at baseline and adapted MIE settings

Months 14-16: Recruitment

Months 16 – 18: Data Collection

N = 50

Testing with OEP for all patients tolerating Work Package 3A and B

IMAGE: Flexible-fiberoptic-laryngoscopy - Fort Worth ENT & Sinus

Months 19-21: Data Analysis
Review of airway status
Comparison of peak cough flows

Work Package 4B - Understand patients', parents' and carers' views, tolerance, and acceptance of new assessment techniques.

Months 16-18: Data Collection

All patients recruited for 3A/B and 4A will partake in semi structured interviews to discuss tolerance of new tests.

- Thematic Analysis
- Key points for relevant for clinical need and population wellbeing identified.

Months 22-36

- Data Synthesis
- Writing Up
- Conclusions
- Recommendations
- Dissemination Events

Appendix #3: Gantt Chart(3).pdf

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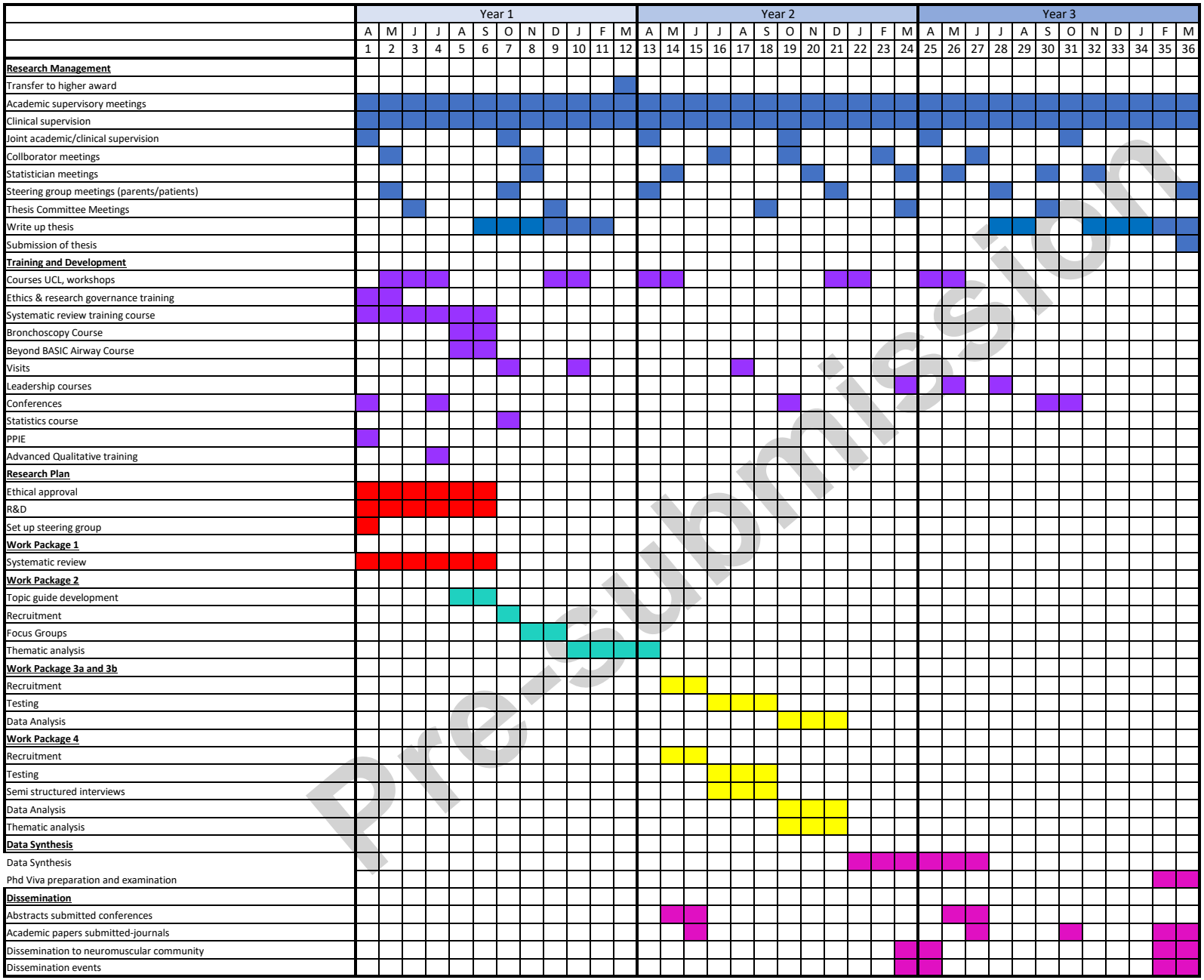

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Appendix #4: 23SH29 - IRAS 34988_SoECAT_TFL for SMA (CPMS 34988) (VALIDATED 24May2024).xlsx

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